

MIRNY STATION, Antarctica

WMO: 895920

Lat: 66.553S

Lon: 93.010E

Elev: 45

StdP: 100.79

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-29.9	-28.0	-34.1	0.2	-29.2	-32.5	0.2	-27.4	43.9	-12.8	38.2	-18.0	8.1	160	1.311

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.8	1.8	-0.1	1.2	-0.6	0.5	-1.1	0.2	1.3	-0.3	0.8	-0.8	0.3	4.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
-0.8	3.6	0.2	-1.4	3.4	-0.1	-2.1	3.2	-0.4	10.3	1.5	9.3	0.9	8.3	0.3	2.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 37.7	31.6	25.7	-32.2	3.5	2.7	1.5	-34.1	4.5	-35.7	5.4	-37.3	6.2	-39.3	7.3		
(5)			-32.3	1.0	2.7	0.8	-34.2	1.6	-35.8	2.1	-37.3	2.6	-39.2	3.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-11.4	-2.3	-5.7	-10.8	-13.9	-15.6	-15.6	-16.4	-16.8	-16.2	-13.7	-7.0	-2.9		
	DBStd	6.90	2.05	2.93	4.00	4.81	5.42	5.43	5.92	5.41	5.49	4.21	3.38	2.10		
	HDD10.0	7818	381	440	643	716	794	768	818	831	785	733	511	399		
	HDD18.3	10860	639	673	902	966	1052	1018	1076	1089	1035	992	761	658		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	8.5	5.6	7.2	7.6	10.2	10.0	11.3	10.4	9.1	8.4	8.3	7.8	6.2		
(15) Precipitation	PrecAvg	537	20	22	36	39	55	66	67	64	65	47	41	22		
	PrecMax	1009	100	63	87	118	194	187	257	190	240	160	197	92		
	PrecMin	161	0	0	0	1	2	6	13	3	3	1	0	0		
	PrecStd	239	21	18	25	31	44	50	55	44	51	38	41	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	1.7	-1.5	-2.3	-2.7	-4.0	-3.0	-3.3	-3.7	-3.8	0.6	2.8		
		MCWB	0.6	-0.2	-2.3	-2.5	-3.2	-4.7	-3.5	-3.7	-4.3	-4.5	-1.7	0.5		
	2%	DB	2.0	0.7	-2.8	-4.3	-5.0	-5.3	-4.7	-5.7	-5.7	-5.1	-0.7	1.7		
		MCWB	0.2	-0.9	-3.5	-4.7	-5.5	-5.8	-5.1	-6.1	-6.2	-5.9	-2.4	-0.2		
	5%	DB	1.4	-0.3	-3.7	-5.7	-6.4	-6.5	-6.4	-7.3	-7.1	-6.3	-1.8	1.0		
		MCWB	-0.4	-1.6	-4.4	-6.4	-6.9	-7.1	-6.7	-7.8	-7.7	-7.1	-3.0	-0.8		
	10%	DB	0.7	-1.2	-4.8	-7.0	-7.9	-8.1	-8.4	-8.9	-8.3	-7.4	-2.4	0.3		
		MCWB	-0.8	-2.5	-5.5	-7.6	-8.4	-8.6	-9.0	-9.4	-8.9	-8.3	-3.5	-1.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	0.2	-2.0	-2.4	-3.2	-4.4	-3.3	-3.6	-4.2	-4.3	-1.3	0.9		
		MCDB	2.3	1.2	-1.6	-2.3	-2.8	-4.2	-3.2	-3.3	-3.9	-4.0	-0.4	2.4		
	2%	WB	0.5	-0.6	-3.4	-4.6	-5.5	-5.7	-5.0	-5.9	-6.1	-5.8	-2.0	0.1		
		MCDB	1.6	0.3	-3.0	-4.3	-5.1	-5.4	-4.7	-5.7	-5.7	-5.3	-1.3	1.3		
	5%	WB	-0.1	-1.5	-4.4	-6.3	-6.8	-7.1	-6.7	-7.7	-7.5	-6.9	-2.7	-0.4		
		MCDB	1.0	-0.4	-3.9	-5.9	-6.4	-6.6	-6.3	-7.3	-7.1	-6.3	-1.9	0.7		
	10%	WB	-0.6	-2.3	-5.5	-7.6	-8.3	-8.6	-8.9	-9.4	-8.9	-8.3	-3.4	-1.0		
		MCDB	0.5	-1.3	-4.8	-7.0	-7.9	-8.1	-8.5	-8.9	-8.4	-7.5	-2.6	0.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.8	5.0	5.0	4.7	4.6	4.6	4.8	5.0	4.9	5.2	4.8	4.1		
		MCDBR	4.0	4.5	4.7	4.5	4.2	4.2	4.5	4.6	3.9	4.3	4.8	4.3		
		MCWBR	3.5	4.3	4.8	4.7	4.3	4.2	4.5	4.6	3.9	4.1	4.2	3.7		
		MCDBR	3.3	4.1	4.6	4.4	4.0	4.0	4.5	4.6	3.9	4.1	3.9	3.6		
	5% WB	MCWBR	3.1	4.1	4.7	4.6	4.1	4.2	4.6	4.7	4.0	4.1	3.7	3.4		
(40) Clear-Sky Solar Irradiance	taub		0.264	0.249	0.227	0.185	0.126	N/A	0.130	0.184	0.215	0.226	0.249	0.264		
	taud		2.347	2.383	2.352	2.203	1.916	N/A	1.887	2.152	2.258	2.265	2.282	2.303		
	Ebn at Noon		998	961	885	712	345	0	294	691	880	979	1008	1011		
	Edh at Noon		107	89	70	42	13	0	12	43	74	99	113	116		
(44) All-Sky Solar Radiation	RadAvg		7.64	5.36	2.85	1.03	0.16	0.00	0.06	0.57	2.10	4.62	7.26	8.59		
	RadStd		0.32	0.27	0.19	0.07	0.01	0.00	0.00	0.03	0.12	0.18	0.35	0.45		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	-0.89	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page